WASP-50b

R-0912

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-50_230916` · created 2026-07-29T06:16:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460203.913 +/- 0.0019 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1368 +/- 0.0094
Transit depth (Rp/Rs)^2	1.87 +/- 0.26 [%]
Orbital Inclination (inc)	84.69 +/- 0.52
Airmass coefficient 1 (a1)	1.2074 +/- 0.0098
Airmass coefficient 2 (a2)	-0.1296 +/- 0.0073
Scatter in the residuals of the lightcurve fit is	0.81 %
Best Comparison Star	None
Optimal Aperture	4.87
Optimal Annulus	7.92
Transit Duration (day)	0.0745 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1368 ± 0.0094 vs 0.1404 (deviation 0.37σ)
Duration fit vs published	0.0745 d vs 0.0754 d (deviation 0.2σ)
Mid-time O–C	-0.6 min
Depth SNR	14.0
Residual scatter	0.81 %

Light curve

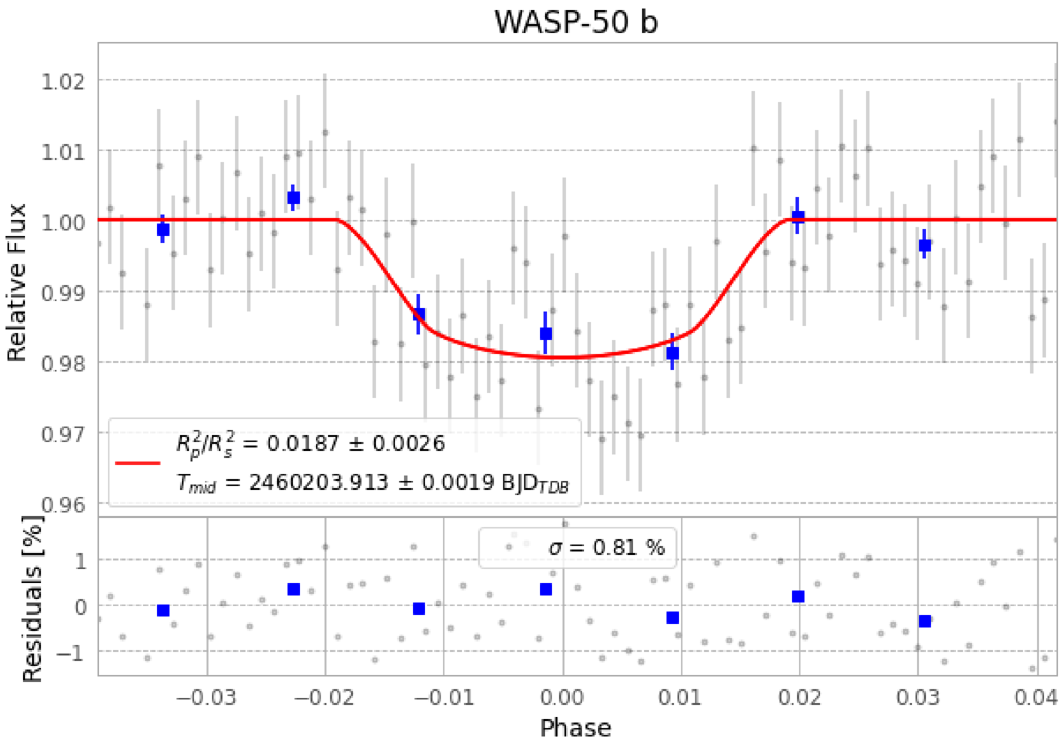

Figure 1 — FinalLightCurve_WASP-50 b_2023-09-16.png (SHA-256 de67d2ffe1b2d7d0…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [358, 214], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 1daf861dd3ac5a8d…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9`

Data provenance

77 FITS frames (51 MB), `WASP-50230916075718.FITS` → `WASP-50230916114515.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-230916.log (63aa7d73d6a1…), FinalLightCurve_WASP-50 b_2023-09-16.png (de67d2ffe1b2…), FinalLightCurve_WASP-50 b_2023-09-16.csv (d48619070b75…)

Compute resources

Wall time 90.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Generated 2026-07-29 06:16Z.